Personal Agent Protocol (PAP / Poppy)

Standardizing AI-to-Business Interactions

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The Problem

The shift

Agents are moving from Q&A to doing complete tasks for you

The dilemma

All or nothing:

Agents are either blocked, or fully signed in as you

The solution

An open standard:

a secure handshake between agents and businesses

Main Overview

What it is

Open, OAuth-based protocol from Meta and Sierra

Timeline

Announced on Oct 6; draft published Oct 9

Core principle

You choose what your agent can access; companies set the limits

Backers

Shopify, Stripe, Walmart + 35 design partners incl. OpenAI, Visa, Mastercard

Step 1: Company-Published Discovery File

It tells agents about the Company's:

- Sign-in options and scopes
- Available APIs or Company Agents
- Authentication endpoints
- Other available capabilities

`/.well-known/poppy.json`

```
{
  "protocol_version": "0.1",
  "organization": {
    "name": "Example Company",
    "domain": "example.com"
  },
  "auth": {
    "issuer": "https://auth.example.com",
    "direct": {
      "scopes": ["poppy:read", "poppy:write", "addresses"]
    },
    "device": {
      "scopes": ["poppy:read", "poppy:write"]
    },
    "mediated": {
      "endpoint": "https://auth.example.com/poppy/sign-in",
      "fields": [
        { "name": "email", "label": "Email", "secret": false },
        { "name": "password", "label": "Password", "secret": true }
      ],
      "scopes": ["poppy:read"]
```

Step 2: Agent Identification

- Agent ID = a URL hosting its public keys
- Sessions start signed out: recognized, but anonymous
- A different user ID per company, so you can't be tracked across businesses
- Short-lived Session Token + long-lived Account Token
- Tokens are tied to the agent's private key, so a stolen token is useless

Step 3: Signing In

Usually on the company's own web page

Direct

Sign in on the company's page (standard OAuth)

Device

Agent shows a link and code; you sign in on your phone

Mediated

Agent signs in with your credentials: the riskier exception

Scopes: what access is granted

```
poppy:read   (view)
```

```
poppy:write   (change)
```

```
custom, e.g. addresses
```

Step 4: Agent Does the Work

Three ways for the agent to work once signed in

APIs

The fast route, using OpenAPI or MCP

Web browsing

The agent proves which session it belongs to, so the site applies your permissions

Conversations

The agent chats with the company's agent; every message is labeled AI or human

Humans can join: the agent can request a live employee, and your agent decides if you're brought in

Takeaways

Users

Get more done faster with their personal agent at every company that supports the protocol.

Sign in directly and control permissions.

Personal agents

More efficient access through the company's APIs or a conversation with its agent, instead of navigating web pages.

Companies

Can connect a website, APIs, or agent, and add more over time.

Understand customer behavior across channels.

Control which agents are allowed and what agents can and can't do.

Only on draft version 0.1 at the moment, no full reference implementation

Sources

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https://sierra.ai/blog/poppy

https://personalagentprotocol.org/

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